

An empirical example: *Rhinoclemmys* data

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I conducted a phylogenetic diversity (PD) analysis for the *Rhinoclemmys* genus to evaluate the effect of branch length on the chosen conservation area(s). The topology used corresponds to a total evidence analysis from (Romero-Alarcon 2020), while the distribution was modified from (Le and McCord 2008).

Read the data: the distribution, and the tree

```
## For reproductibility purposes

set.seed(121)

options(warn=-1)
library(ggplot2)
suppressMessages(library(ggtree))

suppressMessages(library(blepd))

suppressMessages(library(phytools))

## library(ggtree)

##
## If you want to install ggtree.
##
### https://bioconductor.org/packages/release/bioc/html/ggtree.html
##
####
# if (!require("BiocManager", quietly = TRUE))
#   install.packages("BiocManager")
# BiocManager::install("ggtree")

## Version

cat("Analyses made with blepd version:", unlist(packageVersion("blepd"))))

## Analyses made with blepd version: 0 3 7 2025 1 4

##
## The tree file (in Newick format) and the distribution file (in CSV format)
## can be found in the data directory or as a data set within the package.
##
```

```

## Assuming data are read from files, create an object to store the
## distribution and the tree.

RhinoclemmysData <- list()

## Read data

## The distribution is a labeled CSV file with areas as rows and terminals as
## columns.

#setwd("./csv/")

## getwd()

csvFile <- list.files(pattern="csv")

## csvFile

### The functions use a matrix object for the distributions

RhinoclemmysData$distribution <- as.matrix(read.table(csvFile,
  stringsAsFactors=FALSE,
  header=TRUE,
  row.names=1,
  sep=",")
)

print(t(RhinoclemmysData$distribution))

##           Al An-As Cho Nsa Nuh Ph Pl
## R_annulata    1     0  1  0  0  1  0
## R_areolata    1     0  0  0  0  0  0
## R_diademata    0     0  0  1  0  0  0
## R_funerea      1     0  0  0  1  0  0
## R_melanosterna 0     0  1  0  0  1  0
## R_nasuta       0     0  1  0  0  0  0
## R_pulcherrima  0     0  0  0  1  0  1
## R_punctularia  0     1  0  0  0  0  0
## R_rubida       0     0  0  0  0  0  1

##
## Tree(s) in nexus or newick format
##
## setwd("../tree/")
##

treeFiles <- list.files(pattern=".tre")

##treeFiles

RhinoclemmysData$tree <- read.tree(treeFiles)

```

```

## name of tree(s) without .tre

treeFiles <- gsub(".tre","",treeFiles)

##getwd()

## You might need to save the object as an .Rda file.

##
save(RhinoclemmysData, file = "../data/RhinoclemmysData.rda")

## Plotting:

## The tree

## using ggtree

# RhinoclemmysData$tree <- reorder(RhinoclemmysData$tree, order = "cladewise")

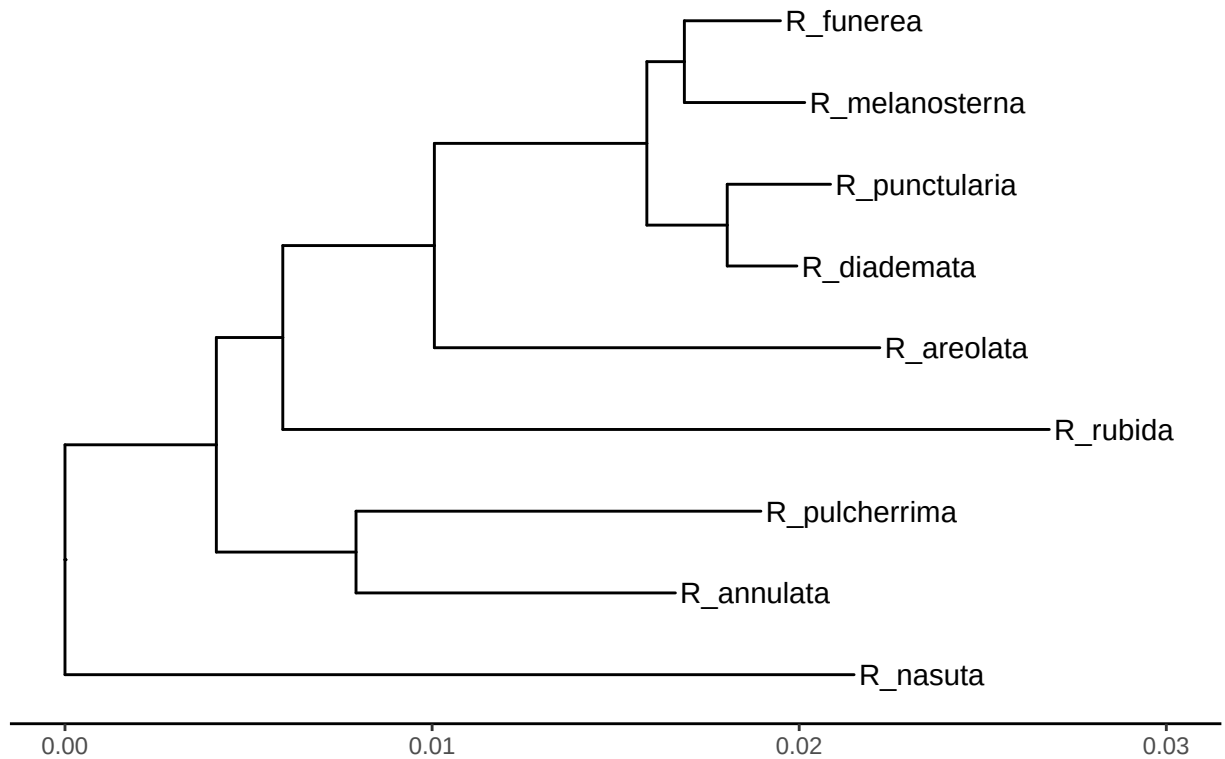
plotTree <- ggtree(RhinoclemmysData$tree, ladderize=TRUE,
                  color="black", size=0.51, linetype="solid") +
  geom_tiplab(size=4, color="black") +
  xlim(0,0.030) +
  theme_tree2() +
  ggtitle(treeFiles[1])

##pdf("RhinoclemmysTopology.pdf", onefile = TRUE)

print(plotTree)

```

Rhinoclemmys_TotalEvidence_ML

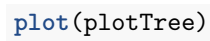


```
##dev.off()

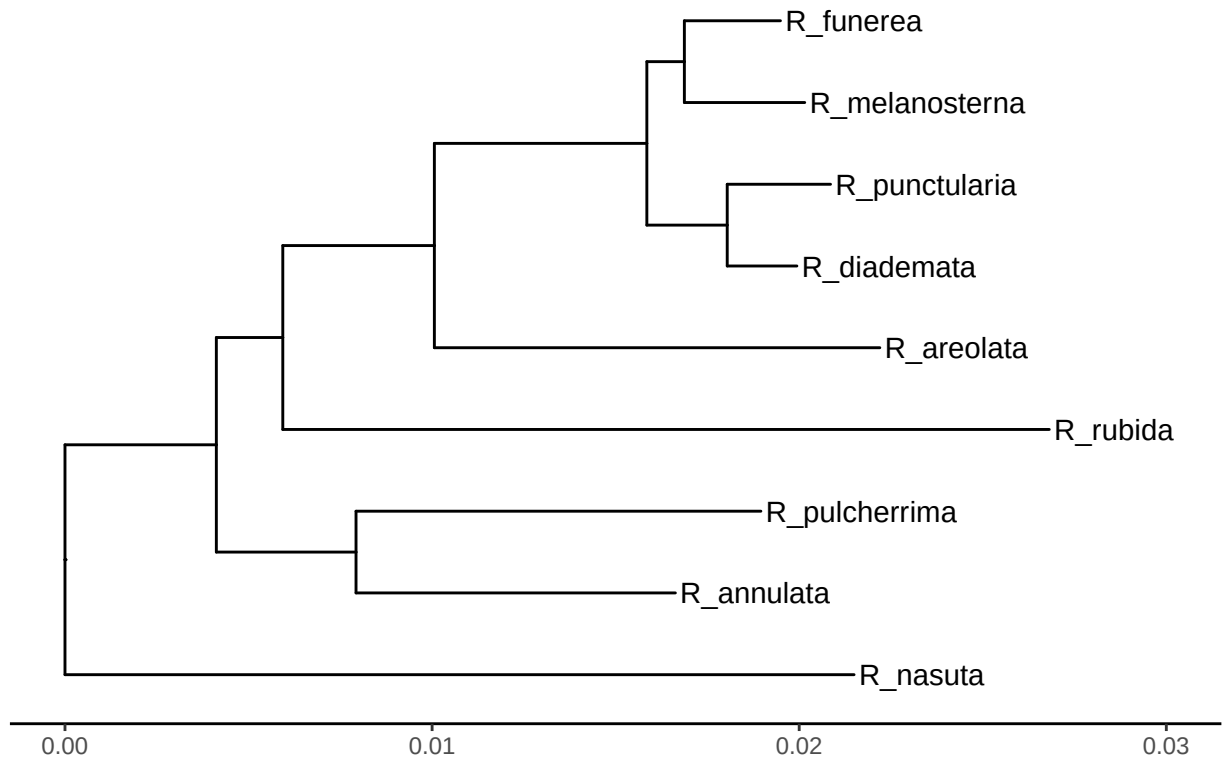
#~ Alternatively, we can plot the tree using APE

plot.phylo(RhinoclemmysData$tree)

# The number of the nodes / tips
#
node.labels()
####
tiplabels()
```



Rhinoclemmys_TotalEvidence_ML



```
print(t(RhinoclemmysData$distribution))
```

```
##           Al An-As Cho Nsa Nuh Ph Pl
## R_annulata 1     0  1  0  0  1  0
## R_areolata 1     0  0  0  0  0  0
## R_diademata 0     0  0  1  0  0  0
## R_funerea  1     0  0  0  1  0  0
## R_melanosterna 0     0  1  0  0  1  0
## R_nasuta   0     0  1  0  0  0  0
## R_pulcherrima 0     0  0  0  1  0  1
## R_punctularia 0     1  0  0  0  0  0
## R_rubida   0     0  0  0  0  0  1
```

```
#Plot the branch lengths.
```

```
library(ggplot2)
```

```
##par(mfrow=c(2,1))
```

```
datos <- RhinoclemmysData$tree$edge.length
```

```
data <- as.data.frame(datos)
```

```
a <- ggplot2::ggplot(data, aes(x=datos)) +
  geom_histogram(bins = 12) +
  labs(title = treeFiles, x = "Branch length: internals and terminals", y = "Frequency") +
  xlim(c(0.001, 0.025)) +
  ylim(c(0, 5))
```

```

terminals <- RhinoclemmysData$tree$edge[,2] < 1 + length(RhinoclemmysData$tree$tip.label)

datos <- RhinoclemmysData$tree$edge.length[terminals]
data <- as.data.frame(datos)

b <- ggplot(data, aes(x=datos)) +
  geom_histogram(bins = 12) +
  labs(x = "Branch length: terminals", y = "Frequency") +
  xlim(c(0.001, 0.025)) +
  ylim(c(0, 5))

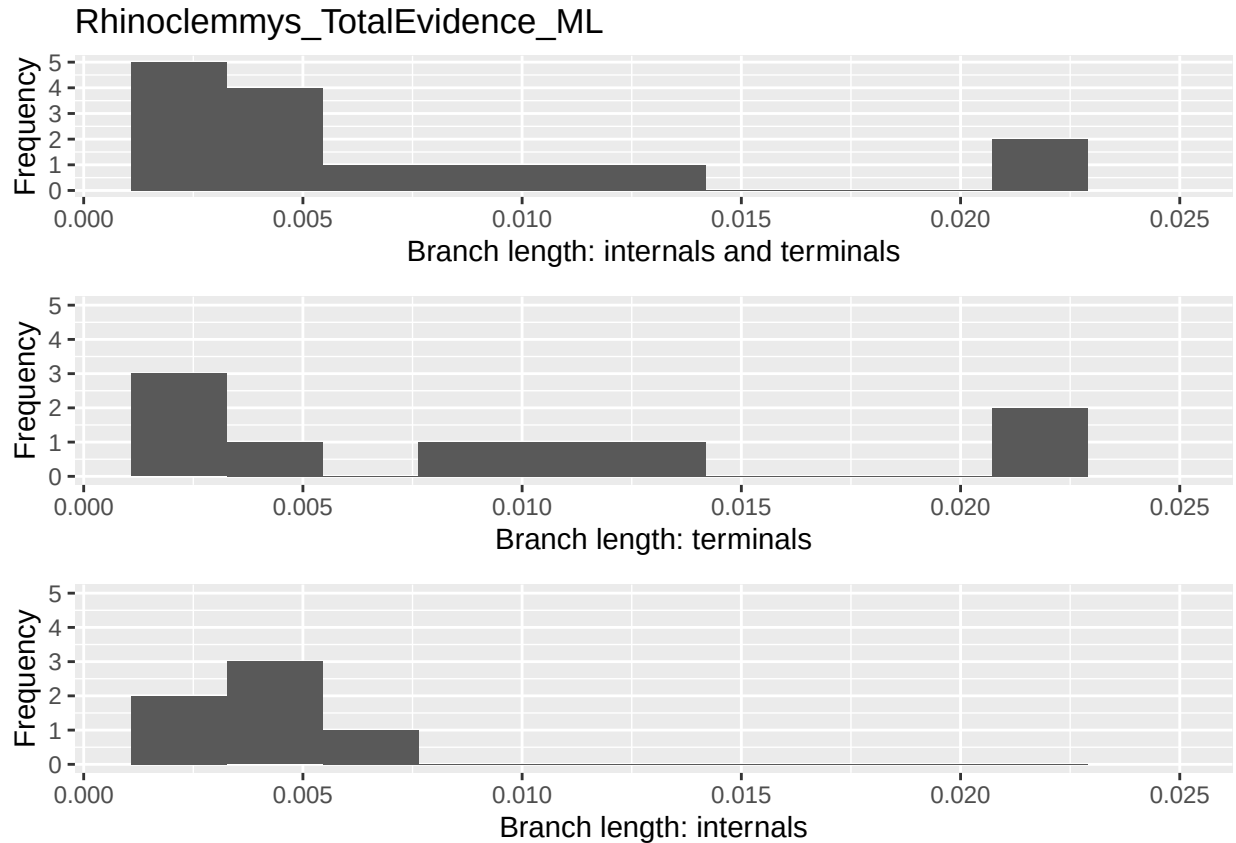
datos <- RhinoclemmysData$tree$edge.length[!terminals]
data <- as.data.frame(datos)

c <- ggplot(data, aes(x=datos)) +
  geom_histogram(bins = 12) +
  labs(x = "Branch length: internals", y = "Frequency") +
  xlim(c(0.001, 0.025)) +
  ylim(c(0, 5))

##pdf("RhinoclemmysBranchLength.pdf", onefile = TRUE)

cowplot::plot_grid(a, b, c, nrow=3)

```



```
##dev.off()

#
# cat("Terminals, with BL larger than 0.02 = ",
# findTerminalgivenLength(RhinoclemmysData$tree,0.02))
#
```

The branch length histograms and the tree plot showed that the internal length branches were similar among each other, and different to terminals'; there were two longer branches (larger than 0.02), *R. nasuta* (inhabiting Cho) and *R. rubida* (Pl), whereas AI and Cho were the richest.

PD values

First, I calculated the PD value for the areas, given this tree.

```
RhinoclemmysData$tablePD <- PDindex(RhinoclemmysData$tree,
                                     distribution=RhinoclemmysData$distribution,
                                     root=TRUE,
                                     percentual = TRUE)

RhinoclemmysData$matrixPD <- as.data.frame(
  matrix(
    unlist(RhinoclemmysData$tablePD),
    nrow= length(treeFiles),
    byrow=TRUE))
```



```

colnames(RhinoclemmysData$matrixPD) <- row.names(RhinoclemmysData$distribution)

row.names(RhinoclemmysData$matrixPD) <- "PD value"

new <- (t(RhinoclemmysData$matrixPD))

values <- new[order(new[,1])]
names <- rownames(new)[order(new[,1])]

print(as.data.frame(paste(names,values)))

```

```

## paste(names, values)
## 1      Nsa 8.05
## 2    An-As 8.42
## 3      Ph 13.18
## 4     Nuh 13.86
## 5      Pl 16.81
## 6      Al 17.82
## 7     Cho 21.86

```

The highest PD was for area Cho, followed by Al, the two richest areas. However, the difference in PD values was not determined by species richness alone, but also by the branch lengths of the species within each region.

#Effect of branch lengths in the PD

I tested the effect of branch length on PD values by swapping terminal and/or internal branch lengths using three different models.

```

for( modelo in c("simpleswap","allswap","uniform") ){

  for( rama in c("terminals","internals","all") ){

    val <- swapBL(RhinoclemmysData$tree,
                  RhinoclemmysData$distribution,
                  model = modelo,
                  branch = rama
                )

    cat( "\n\t Tree=",treeFiles,"\n\t Model=",modelo,
        "\n\t Branchs swapped=",rama,"\n" )

    printSwapBL(val)

    cat( "\n\n" )

  }

}

```

```

## model to test simpleswap reps 100
##
## Tree= Rhinoclemmys_TotalEvidence_ML
## Model= simpleswap

```

```

## Branchs swapped= terminals
## Initial Branch Model A1 Cho P1
## 1 Cho terminals simpleswap 13 85 2
##
##
## model to test simpleswap reps 100
##
## Tree= Rhinoclemmys_TotalEvidence_ML
## Model= simpleswap
## Branchs swapped= internals
## Initial Branch Model Cho
## 1 Cho internals simpleswap 100
##
##
## model to test simpleswap reps 100
##
## Tree= Rhinoclemmys_TotalEvidence_ML
## Model= simpleswap
## Branchs swapped= all
## Initial Branch Model A1 Cho P1
## 1 Cho all simpleswap 8 91 1
##
##
## model to test allswap reps 100
##
## Tree= Rhinoclemmys_TotalEvidence_ML
## Model= allswap
## Branchs swapped= terminals
## Initial Branch Model A1 Cho Nuh P1
## 1 Cho terminals allswap 35 53 10 2
##
##
## model to test allswap reps 100
##
## Tree= Rhinoclemmys_TotalEvidence_ML
## Model= allswap
## Branchs swapped= internals
## Initial Branch Model Cho
## 1 Cho internals allswap 100
##
##
## model to test allswap reps 100
##
## Tree= Rhinoclemmys_TotalEvidence_ML
## Model= allswap
## Branchs swapped= all
## Initial Branch Model A1 An-As Cho Nsa Nuh P1
## 1 Cho all allswap 37 4 33 4 19 3
##
##
## model to test uniform reps 100
##
## Tree= Rhinoclemmys_TotalEvidence_ML
## Model= uniform

```

```

## Branchs swapped= terminals
## Initial Branch Model Al Cho Nuh
## 1 Cho terminals uniform 46 50 4
##
##
## model to test uniform reps 100
##
## Tree= Rhinoclemmys_TotalEvidence_ML
## Model= uniform
## Branchs swapped= internals
## Initial Branch Model Cho
## 1 Cho internals uniform 100
##
##
## model to test uniform reps 100
##
## Tree= Rhinoclemmys_TotalEvidence_ML
## Model= uniform
## Branchs swapped= all
## Initial Branch Model Al An-As Cho Nsa Nuh
## 1 Cho all uniform 38 1 53 1 7

```

I determined that terminal branch lengths were crucial in influencing the chosen area. When I swapped all terminal branch lengths (model = “allswap”) or replaced them with a uniform distribution (model = “uniform”), the optimal area shifted from Cho to Al. Recognizing the uneven distribution of terminal branch lengths, I suspected that the longest branches within the Cho and/or Al areas were driving these results.

I first investigated the impact of the number of replicates on the results.

```

for(repetir in 1:4){

  val <- swapBL( RhinoclemmysData$tree,
                 RhinoclemmysData$distribution,
                 model = "allswap",
                 branch = "all",
                 nTimes = 10**repetir
               )

  printSwapBL(val)
}

```

```

## model to test allswap reps 10
## Initial Branch Model Al An-As Cho Nsa Nuh
## 1 Cho all allswap 40 10 30 10 10
## model to test allswap reps 100
## Initial Branch Model Al An-As Cho Nsa Nuh Pl
## 1 Cho all allswap 26 3 50 4 13 4
## model to test allswap reps 1000
## Initial Branch Model Al An-As Cho Nsa Nuh Pl
## 1 Cho all allswap 37 3.9 39.2 4.6 13.4 1.9
## model to test allswap reps 10000
## Initial Branch Model Al An-As Cho Nsa Nuh Pl
## 1 Cho all allswap 36.55 4.33 38.87 4.43 13.96 1.86

```

I found that results stabilized after approximately 1000 replicates, with most differences arising from using

10 or 100 replicates. As a general guideline, at least 100 replicates are necessary, but 1000 are preferable.

I investigated whether the observed differences in results could be attributed to the lengths of the longest branches.

```
## we can use a single command (*evalBranch*)

AllTerminals <- evalBranch(tree=RhinoclemmysData$tree,
                           distribution=RhinoclemmysData$distribution,
                           branchToEval="terminals",
                           approach="all",
                           printNames=TRUE)

printEvalBranch(AllTerminals)
```

##		node	initialArea	FinalArea	Approach	%Delta
## 1		[R_nasuta]	Cho	Al	lower	-46
## 12		[R_pulcherrima]	Cho	Pl	upper	116
## 13		[R_diademata]	Cho	Nsa	upper	1710
## 14		[R_punctularia]	Cho	An-As	upper	1104
## 16		[R_funerea]	Cho	Al	upper	326
## 17		[R_areolata]	Cho	Al	upper	86
## 18		[R_rubida]	Cho	Pl	upper	59

```
AllInternals <- evalBranch(tree=RhinoclemmysData$tree,
                           distribution=RhinoclemmysData$distribution,
                           branchToEval="internals",
                           approach="all",
                           printNames=TRUE)

printEvalBranch(AllInternals)
```

##		node	initialArea	FinalArea	Approach	%Delta
## 13		[R_diademata/R_punctularia]	Cho	An-As	upper	1695

I found that branch length significantly influenced the selected area. While modifying internal branches had minimal to no effect, altering terminal branch lengths might have an impact on area selection. Specifically, a 46% decrease in the branch length of *R. nasuta* or an 86% increase in the branch length of *R. areolata* shifted the optimal area from Cho to Al. Similarly, a 59% increase in the branch length of *R. rubida* or a 116% increase in the branch length of *R. pulcherrima* resulted in a shift from Cho to Pl.

As an additional exercise, I investigated the impact of ultrametricity. I hypothesized that if the tree were ultrametric, terminal sister pairs would exhibit similar behavior, with differences primarily attributable to the underlying distribution.

```
## now let's see the impact of ultra-metricity

AllTerminalsU <- evalBranch(tree=force.ultrametric(RhinoclemmysData$tree),
                           distribution=RhinoclemmysData$distribution,
                           branchToEval="terminals",
                           approach="all",
                           printNames=TRUE)
```

```
## *****
## *                               Note:                               *
## *   force.ultrametric does not include a formal method to         *
## *   ultrametricize a tree & should only be used to coerce         *
## *   a phylogeny that fails is.ultrametric due to rounding --      *
```

```
## *      not as a substitute for formal rate-smoothing methods.      *
## *****

cat("Ultrametric tree\n")

## Ultrametric tree

printEvalBranch(AllTerminalsU)

##           node initialArea FinalArea Approach %Delta
## 9      [R_nasuta]          Cho        Al    lower   -49
## 11     [R_funerea]          Cho        Al    upper   336
## 12     [R_diademata]        Cho        Nsa    upper  1891
## 13 [R_punctularia]          Cho      An-As    upper  1891
## 14     [R_areolata]          Cho        Al    upper    92
## 15     [R_rubida]           Cho        Pl    upper   135
## 17 [R_pulcherrima]          Cho      NuhPl    upper   210

printEvalBranch(AllTerminals)

##           node initialArea FinalArea Approach %Delta
## 1      [R_nasuta]          Cho        Al    lower   -46
## 12 [R_pulcherrima]          Cho        Pl    upper   116
## 13     [R_diademata]        Cho        Nsa    upper  1710
## 14 [R_punctularia]          Cho      An-As    upper  1104
## 16     [R_funerea]          Cho        Al    upper   326
## 17     [R_areolata]          Cho        Al    upper    86
## 18     [R_rubida]           Cho        Pl    upper    59

AllInternalsU <- evalBranch(tree=force.ultrametric(RhinoclemmysData$tree),
                           distribution=RhinoclemmysData$distribution,
                           branchToEval="internals",
                           approach="all",
                           redondeo=3,
                           printNames=TRUE)

## *****
## *                               Note:                               *
## *      force.ultrametric does not include a formal method to      *
## *      ultrametricize a tree & should only be used to coerce      *
## *      a phylogeny that fails is.ultrametric due to rounding --    *
## *      not as a substitute for formal rate-smoothing methods.      *
## *****

printEvalBranch(AllInternalsU)

##           node initialArea FinalArea Approach %Delta
## 8 [R_diademata/R_punctularia]      Cho An-AsNsa    upper  1860

printEvalBranch(AllInternals)

##           node initialArea FinalArea Approach %Delta
## 13 [R_diademata/R_punctularia]      Cho      An-As    upper  1695
```

These findings suggest that decisions to conserve solely area Cho are not misguided. Areas Al, and potentially Pl, should not be considered as essential components of the conservation range.

Literature cited

- Le, Minh, and William P. McCord. 2008. “Phylogenetic relationships and biogeographical history of the genus *Rhinoclemmys* Fitzinger, 1835 and the monophyly of the turtle family Geoemydidae (Testudines: Testudinoidea).” *Zoological Journal of the Linnean Society* 153 (4): 751–67. <https://doi.org/10.1111/j.1096-3642.2008.00413.x>.
- Romero-Alarcon, L. Viviana. 2020. “A total-evidence phylogeny of the crown and stem-groups of turtles (Pantestudines: Testudinata).” Master’s thesis, Colombia: Escuela de Biología, UIS.